

Determinants

Given four numbers a, b, c , and d , we can visually organize these values in the following ‘grid’:

$$M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

This grid is called a *matrix*. Organized in this fashion, we can define functions on this geometric object. For example, we can define the **determinant** function:

$$\det(M) = \det \left(\begin{bmatrix} a & b \\ c & d \end{bmatrix} \right) = ad - bc.$$

If we have a 3×3 grid, we ‘expand’ via the first row (this is called *cofactor expansion*):

$$\det \left(\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \right) = a \cdot \det \left(\begin{bmatrix} e & f \\ h & i \end{bmatrix} \right) - b \cdot \det \left(\begin{bmatrix} d & f \\ g & i \end{bmatrix} \right) + c \cdot \det \left(\begin{bmatrix} d & e \\ g & h \end{bmatrix} \right)$$

Sometimes, instead of writing ‘det’, we instead replace the ‘box’ around the matrix with straight lines on either side:

$$\det \left(\begin{bmatrix} a & b \\ c & d \end{bmatrix} \right) = \left| \begin{array}{cc} a & b \\ c & d \end{array} \right| = ad - bc$$

Example. Consider the following matrix:

$$M = \begin{bmatrix} 4 & 2 & 3 \\ 3 & 1 & 1 \\ 2 & 1 & 2 \end{bmatrix}.$$

Compute $\det(M)$.

In 2-space, we frequently discuss the topic of how ‘tilted’ a line is – this is the study of *slope*. In 3-space, lines become planes. Thus, how do we determine how ‘tilted’ a plane is? It turns out there is a natural answer: the (unit) **normal** vector **n**.

So how do we compute this vector? We need a process to take a plane’s defining vectors **u** and **v**, and find a vector pointing in the direction of **n**. It turns out, this is the **cross product** of **u** and **v**:

Definition The **cross** (or **vector**) **product** of two vectors **u** and **v** is the vector given by

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} =$$

Given two vectors **u** and **v**, their cross product $\mathbf{u} \times \mathbf{v}$ will be orthogonal to **both** of them. The direction it will face (up or down) obeys the *right-hand rule*: As your fingers curl from **u** to **v**, the cross product $\mathbf{u} \times \mathbf{v}$ will point in the direction of your thumb:

Example. Consider the vectors $\mathbf{u} = \langle 2, 1, 1 \rangle$ and $\mathbf{v} = \langle -4, 3, 1 \rangle$. Compute $\mathbf{u} \times \mathbf{v}$ and $\mathbf{v} \times \mathbf{u}$.

Let us see how this notion plays with the standard unit vectors **i**, **j**, and **k**:

In general, we have the following properties for the cross product: Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ be three vectors and let r and s be scalars.

$$(i) \quad (r\mathbf{u}) \times (s\mathbf{v}) = (rs)(\mathbf{u} \times \mathbf{v})$$

$$(ii) \quad \mathbf{u} \times (\mathbf{v} + \mathbf{w}) = (\mathbf{u} \times \mathbf{v}) + (\mathbf{u} \times \mathbf{w})$$

$$(iii) \quad \mathbf{v} \times \mathbf{u} = -(\mathbf{u} \times \mathbf{v})$$

$$(iv) \quad (\mathbf{v} + \mathbf{w}) \times \mathbf{u} = (\mathbf{v} \times \mathbf{u}) + (\mathbf{w} \times \mathbf{u})$$

$$(v) \quad \vec{0} \times \mathbf{u} = \vec{0}$$

$$(vi) \quad \mathbf{u} \times (\mathbf{v} \times \mathbf{w}) = (\mathbf{u} \cdot \mathbf{w})\mathbf{v} - (\mathbf{u} \cdot \mathbf{v})\mathbf{w}$$

Theorem. Let $\mathbf{u}$ and $\mathbf{v}$ be two vectors and let θ be the angle between them.

$$(i) \quad \mathbf{u} \text{ and } \mathbf{v} \text{ are parallel if and only if } \mathbf{u} \times \mathbf{v} = \vec{0}.$$

$$(ii) \quad \text{The area of the parallelogram with } \mathbf{u} \text{ and } \mathbf{v} \text{ as adjacent sides is equal to } |\mathbf{u} \times \mathbf{v}| = |\mathbf{u}||\mathbf{v}|\sin(\theta).$$

Theorem. Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ be vectors. The quantity $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}$ is called the *triple scalar product*. The volume of the parallelepiped determined by $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ is equal to

$$|(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}| = \left| \det \begin{pmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{pmatrix} \right|$$

Example. Find the volume of the parallelepiped determined by $\mathbf{u} = \langle 1, 2, -1 \rangle$, $\mathbf{v} = \langle -2, 0, 3 \rangle$, and $\mathbf{w} = \langle 0, 7, -4 \rangle$.

An Alternate Way to Compute the Cross-Product: The Fishnet

There is a ‘faster’ way to compute the cross product $\mathbf{u} \times \mathbf{v}$:

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} =$$

Example. Consider the points $P(1, -1, 0)$, $Q(2, 1, -1)$, and $R(-1, 1, 2)$.

(i) Find a unit vector perpendicular to the plane defined by P , Q , and R .

(ii) Find the area of the triangle $\triangle PQR$.